Orthogonal Subspace Unmixing to Address Spectral Variability for Hyperspectral Image

Longfei Ren, Danfeng Hong, *Senior Member, IEEE*, Lianru Gao, *Senior Member, IEEE*, Xu Sun, *Member, IEEE*, Min Huang, and Jocelyn Chanussot, *Fellow, IEEE*

***Abstract*— Hyperspectral unmixing aims at estimating pure spectral signatures and their proportions in each pixel. In practice, the atmospheric effects, intrinsic variation of the spectral signatures of the materials, illumination, and topographic changes cause what is known as spectral variability resulting in significant estimation errors being propagated throughout the unmixing task. To this end, we developed a new method, called the orthogonal subspace unmixing (OSU), to address spectral variability by utilizing the orthogonal subspace projection. The proposed OSU method jointly performs orthogonal subspace learning and the unmixing process to find a more suitable subspace for unmixing. The orthogonal subspace projection encourages the representation held in the subspace to be more distinct from each other to remove the complex spectral variability in the subspace. Furthermore, an alternating minimization (AM) was designed to solve the resulting optimization problem. An efficient and convergent symmetric Gauss–Seidel alternating direction method of multipliers (sGS-ADMM), essentially a special case of the semiproximal alternating direction method of multipliers (SPADMM), was developed to solve the subproblem. Experiments conducted on one synthetic data and two real data demonstrate the effectiveness and superiority of the proposed framework in mitigating the effects of spectral variability with respect to classical linear unmixing methods or variability accounting approaches.**

***Index Terms*— Alternating minimization (AM), hyperspectral image, semiproximal alternating direction method of multipliers (SPADMM), spectral variability, subspace unmixing.**

I. Introduction

HYPERSPECTRAL imaging is a landmark technique in remote sensing, which acquires reflected light typically in hundreds of narrow and contiguous wavelengths of the electromagnetic spectrum, encapsulating more spectral information about the different materials present in the hyperspectral images [1], [2]. The high spectral resolution of hyperspectral images enables them to be a powerful tool for different fields and is now at the core of a large number of applications, such as agriculture, mineralogy, disease diagnosis, and space exploration [3], [4]. However, this high spectral resolution brings about a tradeoff of significantly reduced spatial resolution, resulting in the observed signature being a mixture of several materials [5], [6].

Spectral unmixing is a source separation technique in hyperspectral image exploitation, whose goal is to extract the pure spectral signatures (or endmembers) and to simultaneously estimate their proportions (or abundances) in each pixel [7]. More precisely, the typical spectral unmixing can be decomposed into three stages: 1) estimating the number of endmembers; 2) extracting the endmembers; and 3) quantifying their abundances. As a result of its simplicity, the linear mixing model (LMM), which assumes that the spectral response of a pixel is a linear combination of several endmembers, has attracted huge attention. Despite a large number of LMM-based unmixing methods developed in the literature, one main inherent limitation is that the endmember spectra spatially and spectrally cause what is known as spectral variability leading to inaccurate abundance estimation.

Spectral variability can be caused by atmospheric effects, the intrinsic variation of the spectral signatures of the materials, the illumination, and topographic changes [3]. Recently, enormous efforts modeling estimation error either from spectral bundle- or model-based approaches have been made. The spectral bundle-based approaches address the spectral variability by building a candidate spectral library and trying to model each endmember by a bundle of spectra in the library. Endmember bundles can be collected from field campaigns or can be extracted from the data by using the existing endmember extraction algorithms (EEAs) [7], e.g., the vertex component analysis (VCA) [8], the N-finder [9], or the minimum volume simplex analysis (MVSA) [10]. The endmember bundles contain various instances of each endmember; in this case, unmixing becomes generally equivalent to finding the best signatures in the endmember bundles to represent each pixel in the scenario. Based on the LMM incorporating spectral bundles, multiple-endmember spectral mixture analysis (MESMA) [11] and sparse regression (SR)-based methods [12], [13], [14], [15], [16], [17], [18] are two prevalent manners to address spectral variability problems in hyperspectral unmixing.

Manuscript received 12 December 2022; revised 1 January 2023; accepted 10 January 2023. Date of publication 12 January 2023; date of current version 23 January 2023. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFB3900502; in part by the National Natural Science Foundation of China under Grant 62161160336, Grant 42271350, and Grant 61976226; in part by the MIAI@Grenoble Alpes under Grant ANR-19-P3IA-0003; and in part by the AXA Research Fund. *(Corresponding author: Lianru Gao.)*

Longfei Ren, Danfeng Hong, Lianru Gao, and Xu Sun are with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: renlongfei0131@163.com; hongdf@aircas.ac.cn; gaolr@aircas.ac.cn; sunxu@aircas.ac.cn).

Min Huang is with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the School of Optoelectronics, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: huangmin@aircas.ac.cn).

Jocelyn Chanussot is with the CNRS, Grenoble INP, GIPSA-Lab, University of Grenoble Alpes, 38000 Grenoble, France, and also with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: jocelyn@hi.is).

Digital Object Identifier 10.1109/TGRS.2023.3236471

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:53:14 UTC from IEEE Xplore. Restrictions apply.

Although these approaches are usually easy to interpret and conceptually simple, the accuracy of the unmixing results strongly relies on the quality of the candidate spectral library. The model-based approaches are another physically reasonable and flexible way to handle spectral variability by using statistical or physical information. These strategies provide great freedom to merge prior information and constraints from the underlying application. On the one hand, the statistical model assumes that the representation of endmember spectra is a multivariate distribution, e.g., the beta compositional model [19] or the normal compositional model [20]. On the other hand, the physical-inspired model defines the specific model for the variation of the endmembers to address the spectral variability. In [21], the perturbed LMM (PLMM) explicitly models the endmember variability as an additive perturbation to capture endmember spatial and spectral variability within a given image. In [22], the extended LMM (ELMM) allows a pixelwise variation of each endmember according to scaling factors. In [23], the augmented LMM (ALMM) simultaneously models the scaling factors and other spectral variabilities by means of the endmember dictionary and the additional spectral variability dictionary. In [24], a dynamical framework that models the spectral variability of sources by a scale change and an additive noise term has been developed for multitemporal hyperspectral images. Recent work in [25] has provided a different idea to account for spectral variabilities by using the low-rank learning strategy. The subspace unmixing with low-rank attribute embedding (SULoRA) performs the unmixing in the low-rank subspace instead of the original space, achieving remarkable improvement.

Motivated by the excellent results in [25], we proposed a new orthogonal subspace unmixing (OSU) model to mitigate the effects of spectral variability, without assuming any special fabric to depict the endmember signatures. The proposed OSU model jointly performs the orthogonal subspace learning and unmixing in a closed loop. The orthogonal subspace projection encourages the representation held in the subspace to be more distinct from each other to help keep the complex spectral variability effects out of the subspace. One major enrichment of the proposed model is that it makes less regularization of the subspace projection. On the other hand, it is challenging to solve the OSU model efficiently due to the nonlinear and nonconvex orthogonal constraint, which may lead to many different local minimizers as solutions. In this article, we design a new approach, which is called the alternating minimization (AM) algorithm to solve the model. An efficient and convergent symmetric Gauss–Seidel alternating direction method of multipliers (sGS-ADMM), essentially a special case of the semiproximal alternating direction method of multipliers (SPADMM), is designed to solve the subproblem. According to our experience, the proposed algorithm has the following advantages. First, the orthogonality constraints can be solved analytically, and thus, the algorithm is quite simple and easy to code. Second, the idea of the proposed algorithm can be easily applied to other models as long as the corresponding subproblem can be efficiently solved. More specifically, the main contributions of the work can be summarized as follows.

1) We proposed a new OSU model to account for the complex spectral variabilities by means of the orthogonal subspace projection.
2) To the best of our knowledge, this is the first time to employ the orthogonal subspace projection in the unmixing model. Different from the traditional models that conduct the unmixing in the original space, the derived OSU model conducts the unmixing task in the orthogonal subspace, which broadly mitigates the effects of spectral variability.
3) We developed an AM algorithm to optimize the proposed model, significantly improving unmixing performance without sacrificing computational efficiency.

The rest of this article is organized as follows. Section II reviews the related work. Section III describes the proposed approach and solving algorithm. Section IV gives experimental results obtained on both synthetic data and real data. The conclusion is drawn in Section V. The Appendix gives the proof of Theorem 1.

II. Related Work

In this section, we first depict the LMM and then review several related state-of-the-art mixing models against spectral variability while focusing on their advantages and disadvantages.

A. Conventional LMM

The basic model to represent the interaction between the endmembers and the light is the LMM, which assumes that the spectral response of a pixel is a linear combination of several endmembers presented in the pixel. For each pixel, the conventional LMM can be written as [7]:

$$\mathbf{y}_i = \mathbf{A}\mathbf{s}_i + \mathbf{n}_i \tag{1}$$

where $\mathbf{y}_i := [\mathbf{y}_{i,1}, \mathbf{y}_{i,2}, \ldots, \mathbf{y}_{i,L}]^T \in \mathbb{R}^{L\times 1}$ denotes the measured value of the reflectance; $\mathbf{A} := [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_m] \in \mathbb{R}^{L\times m}$ denotes the endmember matrix containing m endmembers and each $\mathbf{a}_j := [\mathbf{a}_{j,1}, \mathbf{a}_{j,2}, \ldots, \mathbf{a}_{j,L}]^T \in \mathbb{R}^{L\times 1}$ $(j = 1, 2, \ldots, m)$ denotes the L-spectrum of the jth endmember; $\mathbf{s}_i := [\mathbf{s}_{i,1}, \mathbf{s}_{i,2}, \ldots, \mathbf{s}_{i,m}]^T \in \mathbb{R}^{m\times 1}$ denotes the abundances associated with the endmembers; and $\mathbf{n}_i := [\mathbf{n}_{i,1}, \mathbf{n}_{i,2}, \ldots, \mathbf{n}_{i,L}] \in \mathbb{R}^{L\times 1}$ denotes an additive noise vector. The abundances usually satisfy the so-called abundance nonnegative constraint (ANC) and the abundance sum-to-one constraint (ASC). Considering all pixels, the matrix form of LMM can be formulated as

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{N} \tag{2}$$

where $\mathbf{Y} := [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_n] \in \mathbb{R}^{L\times n}$, $\mathbf{X} := [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n] \in \mathbb{R}^{m\times n}$, and $\mathbf{N} := [\mathbf{n}_1, \mathbf{n}_2, \ldots, \mathbf{n}_n] \in \mathbb{R}^{L\times n}$ denote the whole observed hyperspectral image, the corresponding abundances matrix, and the noise matrix, respectively.

Traditionally, the LMM assumes that each endmember in the given scenarios can be represented by a single signature. Although this assumption brings a computationally simpler and well-posed framework, it stints the applicability of the LMM to address the spectral variability for unmixing task.

B. LMM Incorporating Spectral Bundles

Spectral bundles are an efficient manner to consider spectral variability by building a candidate spectral library. A common approach to building the spectral library is by extracting

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:53:14 UTC from IEEE Xplore. Restrictions apply.

spectra directly from the observed images, or using libraries known in advance, such as the publicly available United States Geological Survey (USGS) digital library,[1] which contains over 1300 mineral signatures. The building spectral library sometimes called bundles, in principle, should account for all possible variations of each endmember, which can be written as [3]

$$\mathbf{M}_p = \{\mathbf{m}_{p,1}, \mathbf{m}_{p,2}, \ldots, \mathbf{m}_{p,M_p}\}, \quad p = 1, \ldots, P \tag{3}$$

where $\mathbf{M}_p$ denotes a bundle containing M_p spectral signature $\mathbf{m}_{p,i}$ of the pth material and P denotes the number of endmembers. Using the building endmember library, the LMM incorporating spectral bundles can be written as [14]

$$\mathbf{Y} = \mathbf{M}\mathbf{X} + \mathbf{N} \tag{4}$$

where the endmember matrix $\mathbf{A}$ in (7) is replaced by a known spectral library $\mathbf{M}$

$$\mathbf{M} = [\mathbf{M}_1, \ldots, \mathbf{M}_P]. \tag{5}$$

In the following, two classical LMM incorporating spectral bundle-based unmixing methods are introduced.

1) MESMA [11]: The MESMA method enables the endmembers to vary on each pixel basis while following (3). The unmixing issue is solved by finding the combinations of the endmember and abundance that acquire the smallest reconstruction error for each pixel. Although this method generally provides good behavior, it is quite computationally expensive because of the need to test a large number of combinations of endmember, making it unrealistic for large spectral library sizes.

2) SR: To circumvent the high computation cost, SR-based methods have been developed to exploit the structure of the spectral bundles. Unmixing, in this case, becomes generally equivalent to finding the best signatures in the bundles to represent each pixel in the scenario. In [12] and [26], the sparse unmixing via variable splitting and augmented Lagrangian (SUnSAL) and collaborative SUnSAL (CLSUnSAL) are the most common unmixing models, which exploit the single sparsity or the global row sparsity of the abundance coefficient. To account for the spatial information, the total variation (TV) [27], [28], [29], [30], [31] and low-rank [32], [33], [34], [35] regularizers have been widely embedded in the unmixing model. In [17] and [14], the social sparsity and the collaborative hierarchical sparsity are developed to more refinery exploit the structure of the spectral bundles. The global abundance maps can be acquired simply by summing the contributions of all candidates in each bundle. Although these methods are computationally efficient, the unmixing results may not be as accurate as MESMA. Moreover, the process may be troublesome to interpret, e.g., it may pick multiple signatures of an identical material to represent a given pixel.

C. Model-Based Approaches

The model-based approaches are another promising alternative to address spectral variability by defining the specific model for the variation of the endmembers, be it more physics-inspired or computational. Intuitively, the model-based approaches can be generalized into a unified model as [36]

$$\mathbf{Y} = f(\mathbf{A}, \mathbf{X}) + \mathbf{N} \tag{6}$$

where f denotes a linear function to model the spectral variability with respect to the endmember matrix $\mathbf{A}$ and the abundances $\mathbf{X}$.

In the following, we detail four state-of-the-art model-based unmixing methods.

1) PLMM [21]:: In the absence of any prior information on the nature of variability, the PLMM attempts to model the spectral variability by using an additive perturbation of some reference endmembers, which is formulated by

$$f(\mathbf{A}, \mathbf{X}) = \mathbf{A}\mathbf{X} + \underbrace{[\Delta_1 \mathbf{x}_1 | \cdots | \Delta_k \mathbf{x}_k | \cdots | \Delta_n \mathbf{x}_n |]}_{\Delta} \tag{7}$$

where $\Delta_k \in \mathbb{R}^{m \times n}$ denotes the perturbation endmember matrix, whose columns are the perturbation vectors for the kth pixel. The PLMM can overcome the effects of spectral variabilities to a certain extent, While the principal spectral variability such as scaling factors, cannot be merely represented by an additional term.

2) ELMM [22]: It allows a pixelwise variation of each endmember, which adjusts endmembers by multiplying scaling factors to imbibe illumination and contrast effects. This model is formulated by

$$f(\mathbf{A}, \mathbf{X}) = \mathbf{A}(\mathbf{S} \circ \mathbf{X}) \tag{8}$$

where $\mathbf{S} \in \mathbb{R}^{m \times n}$ aims at gathering all the scaling factors for all endmembers and all pixels and $\circ$ denotes the Schur–Hadamard product. The ELMM mainly concentrates on characterizing the scaling factors, but a slight deficiency is ignoring more complex spectral variability.

3) ALMM [23]: To overcome the limitations of the PLMM and the ELMM, the ALMM simultaneously models the scaling factors and other spectral variabilities according to their distinctive properties, which is expressed by

$$f(\mathbf{A}, \mathbf{X}) = \mathbf{A}\mathbf{X}\mathbf{S} + \mathbf{E}\mathbf{B} \tag{9}$$

where $\mathbf{E} \in \mathbb{R}^{L \times r}$ denotes the spectral variability matrix, $\mathbf{B} \in \mathbb{R}^{r \times n}$ denotes the coefficient matrix corresponding to $\mathbf{E}$, and r denotes the number of basis vector in $\mathbf{E}$. The ALMM characters the spectral signature by the endmember dictionary term $\mathbf{AXS}$ with pixelwise scaling and also other spectral variabilities using scaling by the spectral variability term $\mathbf{EB}$. Although the explicit physical meaning of ALMM, some degree of expert information from the endmember is required to determine good unmixing performance.

4) SULoRA [25]: The SULoRA provides a new insight to mitigate the influence of spectral variability. The subspace-based model conducts the unmixing in the robust subspace instead of the original space, which can be expressed by

$$f(\mathbf{A}, \mathbf{X}) = \Theta\mathbf{A}\mathbf{X} \tag{10}$$

where $\Theta \in \mathbb{R}^{L \times L}$ denotes the projection operator to project the original space to the subspace. The SULoRA estimates the subspace projection of the observed hyperspectral image to minimize the influence of spectral variability in the unmixing process, showing significant potential for performance improvement.

[1] Available online: http://speclab.cr.usgs.gov/spectral.lib06

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:53:14 UTC from IEEE Xplore. Restrictions apply.

III. Proposed Approach and Solving Algorithm

A. Model Formulation

Before stepping into the proposed OSU approach, we further introduce the SULoRA as it is the cornerstone of our approach. The SULoRA performs the unmixing in a raw subspace rather than in the original space, more precisely, by minimizing the reconstruction error in the projected space. In the learned subspace, the SULoRA assumes that the pixels belonging to the same class are strongly correlated; hence, the subspace projection is endowed with a low-rank attribute. Furthermore, this process can be mathematically expressed as [25]

$$\begin{aligned} &\mathbf{Y} = \mathbf{Y}' + \mathbf{N}', \qquad \text{s.t. } \mathbf{Y}' = \mathbf{\Theta Y} \\ &\mathbf{Y}' = \mathbf{\Theta A X} + \mathbf{N}'' \end{aligned} \tag{11}$$

where $\mathbf{Y}'$ denotes the spectral subspace representation of the original hyperspectral image; $\mathbf{N}'$ and $\mathbf{N}''$ denote the noise matrix; and $\mathbf{\Theta}$ denotes the subspace projection, which is constrained to be low-dimensional by penalizing the nuclear norm of the projection operator in the cost function and jointly estimating the abundances during the unmixing process.

Inspired by the SULoRA showing that using the low-rank attribute embedding to conduct the unmixing can remarkably remove the spectral variability, we further improve this strategy by using the orthogonal subspace to make it more applicable to the unmixing task. Step by step, we first expound on the motivation for using the orthogonal subspace to conduct the unmixing task in detail. In practice, spectral variability is an inevitable phenomenon that exists in hyperspectral images, which can be caused, for instance, by atmospheric conditions, topography change, nonlinear material mixing, and variable illumination [3]. These spectral variabilities are complex and quite arduous to accurately characterize with common models and can be treated as an unknown complex noise. Hence, we expect to remove the noise coupled in the image by using the subspace-based method. The subspace representation-based method has been extensively studied in many applications tasks [37], [38], [39], [40], [41]. Among them, the orthogonality, which promotes each base to be more distinct from the other, has been successfully applied for helping keep noise out of the image. Therefore, in this work, we proposed a new framework based on the orthogonal subspace projection to conduct the unmixing. One significant enrichment of the proposed framework is that it provides a parsimonious model without the need for additional regularizations for the projection.

Our proposed OSU model to deal with the complex spectral variability effects can be formulated as the following constrained optimization problem:

$$\begin{aligned} &\underset{\mathbf{\Theta},\mathbf{X}}{\text{argmin}} \; J(\mathbf{\Theta}, \mathbf{X}) := \frac{1}{2}\|\mathbf{\Theta}^T\mathbf{A X} - \mathbf{\Theta}^T\mathbf{Y}\|_F^2 + \Phi(\mathbf{X}) \\ &\text{s.t. } \mathbf{\Theta}^T\mathbf{\Theta} = \mathbf{I}, \quad \mathbf{X} \geq 0 \end{aligned} \tag{12}$$

where the goal is to estimate the projection $\mathbf{\Theta}$ and the abundance matrix $\mathbf{X}$, while the observed hyperspectral image $\mathbf{Y}$ and the endmember matrix $\mathbf{A}$ are given. Similar to the SULoRA, the proposed framework belongs to the semisupervised-based approach, assuming that the endmember matrix $\mathbf{A}$ is known in advance. The number of endmembers in the image is first estimated via HySime [42] and followed by extracting the endmember matrix $\mathbf{A}$ from the image by VCA [8]. Here, the first term of the objective function is the modeling term enforcing the subspace-based model, and the second term of the objective function is the regularization term applying to the abundances. The subspace is forced by the orthogonal constraint, and the abundance is forced by the ANC to satisfy the physical assumption. We should mention that the orthogonal constraint is quite significant here, which encourages the representation held in subspace to be more distinct from each other to help keep the complex spectral variability effects out of the subspace and allows a closed-form solution for computing $\mathbf{\Theta}$. As pointed out in [25], the ASC is not considered in the model since it is too constrained and prone to misestimated abundances. The regularization term is described in the following.

By considering that only a small number of the endmembers are active in each pixel for the real scene, i.e., the abundances corresponding to the endmember matrix are sparse, we choose the $l_{1,1}$ norm as a sparse regularizer. To further exploit the spatial information in the hyperspectral image, we also adopt the TV norm to promote spatial homogeneity and the smoothness between adjacent pixels. The final resulting expression of the regularization with respect to $\mathbf{X}$ can be expressed as

$$\Phi(\mathbf{X}) = \alpha\|\mathbf{X}\|_{1,1} + \beta\|\mathcal{H}\mathbf{X}\|_{1,1} \tag{13}$$

where α and β are regularization parameters, $\|\mathbf{X}\|_{1,1} := \sum_{j=1}^{n}\sum_{i=1}^{m}|\mathbf{X}_{i,j}|$ with $\mathbf{X}_{i,j}$ being an element in the ith row jth column of abundances matrix $\mathbf{X}$, and $\mathcal{H}$ denotes the linear operator of horizontal and vertical gradients between adjacent pixels. Note that the same regularization term penalized on the abundance is also used by SUnSAL-TV [27]; however, one major difference is that our OSU model conducts the unmixing in the projected subspace, while SUnSAL-TV does it in the original space. In the literature, two prevalent manners have been widely investigated to handle the TV norm. On the one hand, the TV norm has been assumed to have periodic boundaries [27], [31], and then, the fast Fourier transform is adopted to handle it. On the other hand, the TV norm has been assumed to have reflexive boundaries [28], [30], [43] and the fast projected gradient-based algorithm is adopted to handle it from a dual perspective. Recent work in [29] has brought new insight into handling the TV norm. The authors assume that the TV norm has reflexive boundaries and split it into the horizontal and vertical TV norms, which are then computed efficiently using the 1-D TV denoising algorithm [44]. In this work, we extend the approach in [29] to solve the problem (12).

B. AM Algorithm

It is challenging to solve the OSU model efficiently due to the nonlinear and nonconvex orthogonality constraint. In this article, we propose to estimate the projection and the abundance matrix alternately iteratively by the AM strategy until the stopping criterion is satisfied, as shown in Algorithm 1. As will be shown in the sequel, the problem has the closed-form solution with respect to $\mathbf{\Theta}$ and hence can be solved very effectively. While the problem is convex with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:53:14 UTC from IEEE Xplore. Restrictions apply.

respect to $\mathbf{X}$, an efficient and convergent sGS-ADMM [45] is well-designed to solve the subproblem.

Algorithm 1 AM Scheme to Find a Local Minimum of (12)

Require: Select an initial point $(\mathbf{\Theta}^0, \mathbf{X}^0)$. Set $k = 0$. Iterate the following steps until the stopping criterion is satisfied:

Step 1. Compute:

$$\mathbf{\Theta}^{k+1} \leftarrow \underset{\mathbf{\Theta}^T\mathbf{\Theta}=\mathbf{I}}{\operatorname{argmin}} J(\mathbf{\Theta}, \mathbf{X}^k).$$

Step 2. Compute:

$$\mathbf{X}^{k+1} \leftarrow \underset{\mathbf{X}\geq 0}{\operatorname{argmin}} J(\mathbf{\Theta}^{k+1}, \mathbf{X}).$$

Step 3. Update:

$$k \leftarrow k + 1.$$

1) Optimization With Respect to $\mathbf{\Theta}$*:* The optimization problem with respect to $\mathbf{\Theta}$ can be expressed as

$$\begin{aligned} &\underset{\mathbf{\Theta}}{\operatorname{argmin}} \ \frac{1}{2}\|\mathbf{\Theta}^T\mathbf{A}\mathbf{X} - \mathbf{\Theta}^T\mathbf{Y}\|_F^2 \\ &\text{s.t. } \mathbf{\Theta}^T\mathbf{\Theta} = \mathbf{I}. \end{aligned} \tag{14}$$

By some algebra, (14) is equivalent to the following problem:

$$\begin{aligned} &\underset{\mathbf{\Theta}}{\operatorname{argmin}} \ \operatorname{tr}\left\{\mathbf{\Theta}^T\left[\frac{1}{2}(\mathbf{A}\mathbf{X} - \mathbf{Y})(\mathbf{A}\mathbf{X} - \mathbf{Y})^T\right]\mathbf{\Theta}\right\} \\ &\text{s.t. } \mathbf{\Theta}^T\mathbf{\Theta} = \mathbf{I}. \end{aligned} \tag{15}$$

Thus, the optimal $\mathbf{\Theta}$ can be obtained by computing eigenvectors of the matrix $(1/2)(\mathbf{A}\mathbf{X} - \mathbf{Y})(\mathbf{A}\mathbf{X} - \mathbf{Y})^T$.

2) Optimization With Respect to $\mathbf{X}$*:* The optimization problem with respect to $\mathbf{X}$ can be expressed as

$$\begin{aligned} &\underset{\mathbf{X}}{\operatorname{argmin}} \ \frac{1}{2}\|\mathbf{\Theta}^T\mathbf{A}\mathbf{X} - \mathbf{\Theta}^T\mathbf{Y}\|_F^2 + \Phi(\mathbf{X}) \\ &\text{s.t. } \mathbf{X} \geq 0. \end{aligned} \tag{16}$$

An equivalent form of (16) can be expressed as

$$\begin{aligned} \underset{\mathbf{X}}{\operatorname{argmin}} \ &\frac{1}{2}\|\mathbf{\Theta}^T\mathbf{A}\mathbf{X} - \mathbf{\Theta}^T\mathbf{Y}\|_F^2 + \alpha\|\mathbf{X}\|_{1,1} + \beta\|\mathcal{H}_h\mathbf{X}\|_{1,1} \\ &+\beta\|\mathcal{H}_v\mathbf{X}\|_{1,1} + \delta_{\mathbb{R}_+^{m\times n}}(\mathbf{X}) \end{aligned} \tag{17}$$

where $\mathcal{H}_h$ and $\mathcal{H}_v$ denote the linear operator of horizontal and vertical gradients between adjacent pixels, respectively; and $\delta_{\mathbb{R}_+^{m\times n}}(\mathbf{X})$ denotes the indicator function, i.e., $\delta_{\mathbb{R}_+^{m\times n}}(\mathbf{X}) = 0$ if $\mathbf{X} \geq 0$ and $\delta_{\mathbb{R}_+^{m\times n}}(\mathbf{X}) = \infty$ if $\mathbf{X} < 0$.

If $p(\mathbf{X}) = \alpha\|\mathbf{X}\|_{1,1} + \beta\|\mathcal{H}_v\mathbf{X}\|_{1,1} + \delta_{\mathbb{R}_+^{m\times n}}(\mathbf{X})$ and $q(\mathbf{X}) = \beta\|\mathcal{H}_h\mathbf{X}\|_{1,1}$, (17) can be rewritten in a simple form

$$\underset{\mathbf{X}}{\operatorname{argmin}} \ \frac{1}{2}\|\mathbf{\Theta}^T\mathbf{A}\mathbf{X} - \mathbf{\Theta}^T\mathbf{Y}\|_F^2 + p(\mathbf{X}) + q(\mathbf{X}). \tag{18}$$

By introducing three slack variables $\mathbf{U}_1 \in \mathbb{R}^{m\times n}$, $\mathbf{U}_2 \in \mathbb{R}^{m\times n}$, and $\mathbf{U}_3 \in \mathbb{R}^{L\times n}$, (18) can be rewritten as

$$\begin{aligned} &\underset{\mathbf{X},\mathbf{U}_1,\mathbf{U}_2,\mathbf{U}_3}{\operatorname{argmin}} \ \frac{1}{2}\|\mathbf{U}_3\|_F^2 + p(\mathbf{U}_1) + q(\mathbf{U}_2) \\ &\text{s.t. } \mathbf{X} = \mathbf{U}_1, \ \mathbf{X} = \mathbf{U}_2, \ \mathbf{\Theta}^T\mathbf{A}\mathbf{X} - \mathbf{\Theta}^T\mathbf{Y} = \mathbf{U}_3. \end{aligned} \tag{19}$$

Algorithm 2 sGS-ADMM

Require: Select an initial point $\left(\mathbf{V}_1^0, \mathbf{V}_2^0, \mathbf{V}_3^0; \mathbf{X}^0\right)$. Set $j = 0$, choose $\sigma > 0$, $\tau \in \left(0, \frac{1+\sqrt{5}}{2}\right)$ and $\{\tilde{\epsilon}_j\}_{j\geq 0}$ be a summable sequence of nonnegative numbers. Iterate the following steps until the stopping criterion is satisfied:

Step 1. Compute:

$$\begin{aligned} \hat{\mathbf{V}}_3^j &\leftarrow \underset{\mathbf{V}_3}{\operatorname{argmin}}\{L_\sigma(\mathbf{V}_1^j, \mathbf{V}_2^j, \mathbf{V}_3; \mathbf{X}^j) - \langle\hat{\delta}_{\mathbf{V}_3}^j, \mathbf{V}_3\rangle\}, \\ \mathbf{V}_1^{j+1} &\leftarrow \underset{\mathbf{V}_1}{\operatorname{argmin}}\{L_\sigma(\mathbf{V}_1, \mathbf{V}_2^j, \hat{\mathbf{V}}_3^j; \mathbf{X}^j)\}, \\ \mathbf{V}_3^{j+1} &\leftarrow \underset{\mathbf{V}_3}{\operatorname{argmin}}\{L_\sigma(\mathbf{V}_1^{j+1}, \mathbf{V}_2^j, \mathbf{V}_3; \mathbf{X}^j) - \langle\delta_{\mathbf{V}_3}^j, \mathbf{V}_3\rangle\}, \\ \mathbf{V}_2^{j+1} &\leftarrow \underset{\mathbf{V}_2}{\operatorname{argmin}}\{L_\sigma(\mathbf{V}_1^{j+1}, \mathbf{V}_2, \mathbf{V}_3^{j+1}; \mathbf{X}^j)\}, \end{aligned}$$

where

$$\begin{aligned} \hat{\delta}_{\mathbf{V}_3}^j &= \nabla_{\mathbf{V}_3}L_\sigma(\mathbf{V}_1^j, \mathbf{V}_2^j, \hat{\mathbf{V}}_3^j; \mathbf{X}^j) \text{ with } \|\hat{\delta}_{\mathbf{V}_3}^j\|_F \leq \tilde{\epsilon}_j, \\ \delta_{\mathbf{V}_3}^j &= \nabla_{\mathbf{V}_3}L_\sigma(\mathbf{V}_1^{j+1}, \mathbf{V}_2^j, \mathbf{V}_3^{j+1}; \mathbf{X}^j) \text{ with } \|\delta_{\mathbf{V}_3}^j\|_F \leq \tilde{\epsilon}_j. \end{aligned}$$

Step 2. Update:

$$\begin{aligned} &\mathbf{X}^{j+1} \leftarrow \mathbf{X}^j - \tau\sigma(-\mathbf{V}_1^{j+1} - \mathbf{V}_2^{j+1} - \mathbf{A}^T\mathbf{V}_3^{j+1}), \\ &j \leftarrow j + 1. \end{aligned}$$

The Lagrangian function of (19) is given by

$$\begin{aligned} &L(\mathbf{X}, \mathbf{U}_1, \mathbf{U}_2, \mathbf{U}_3; \mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3) \\ &= \frac{1}{2}\|\mathbf{U}_3\|_F^2 + p(\mathbf{U}_1) + q(\mathbf{U}_2) \\ &\quad -\langle\mathbf{V}_1, \mathbf{X} - \mathbf{U}_1\rangle - \langle\mathbf{V}_2, \mathbf{X} - \mathbf{U}_2\rangle - \langle\mathbf{V}_3, \mathbf{\Theta}^T\mathbf{A}\mathbf{X} - \mathbf{\Theta}^T\mathbf{Y} - \mathbf{U}_3\rangle \end{aligned} \tag{20}$$

where $\mathbf{V}_1 \in \mathbb{R}^{m\times n}$, $\mathbf{V}_2 \in \mathbb{R}^{m\times n}$, and $\mathbf{V}_3 \in \mathbb{R}^{L\times n}$ are dual variables. Then, we can obtain the dual of the problem (19) from (20) as

$$\begin{aligned} &\underset{\mathbf{V}_1,\mathbf{V}_2,\mathbf{V}_3}{\operatorname{argmin}} \ p^*(-\mathbf{V}_1) + q^*(-\mathbf{V}_2) + \frac{1}{2}\|\mathbf{V}_3\|_F^2 - \left\langle\mathbf{V}_3, \mathbf{\Theta}^T\mathbf{Y}\right\rangle \\ &\text{s.t. } -\mathbf{V}_1 - \mathbf{V}_2 - \mathbf{A}^T\mathbf{\Theta}\mathbf{V}_3 = 0 \end{aligned} \tag{21}$$

where $p^*(\cdot)$ and $q^*(\cdot)$ denote the conjugate function of $p(\cdot)$ and $q(\cdot)$, respectively. For any convex function $p : \mathbb{R}^{m\times n} \to (-\infty, +\infty]$, the conjugate function of p is defined as

$$p^*(\mathbf{Z}) = \sup_{\mathbf{Z}'}\{\operatorname{trace}(\mathbf{Z}^T\mathbf{Z}') - p(\mathbf{Z}')\}. \tag{22}$$

If $\sigma > 0$ is a penalty parameter, the augmented Lagrangian function of problem (21) is given by

$$\begin{aligned} &L_\sigma(\mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3; \mathbf{X}) \\ &= p^*(-\mathbf{V}_1) + q^*(-\mathbf{V}_2) + \frac{1}{2}\|\mathbf{V}_3\|_F^2 - \langle\mathbf{V}_3, \mathbf{\Theta}^T\mathbf{Y}\rangle \\ &\quad +\frac{\sigma}{2}\|-\mathbf{V}_1 - \mathbf{V}_2 - \mathbf{A}^T\mathbf{\Theta}\mathbf{V}_3 - \sigma^{-1}\mathbf{X}\|_F^2 - \frac{1}{2\sigma}\|\mathbf{X}\|_F^2. \end{aligned} \tag{23}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:53:14 UTC from IEEE Xplore. Restrictions apply.

Suppose that $\hat{\delta}_{\mathbf{V}_3}$ and $\delta_{\mathbf{V}_3}$ are given tolerance vectors. The proposed sGS-ADMM to optimize the problem with respect to $\mathbf{X}$ is presented in Algorithm 2. The details of Step 1 in Algorithm 2 are addressed as follows. The minimization of (23) with respect to $\mathbf{V}_3$ can be performed as

$$\mathbf{V}_3 = \underset{\mathbf{V}_3}{\operatorname{argmin}} \ \frac{1}{2}\|\mathbf{V}_3\|_F^2 - \langle \mathbf{V}_3, \mathbf{\Theta}^T\mathbf{Y}\rangle + \frac{\sigma}{2}\| - \mathbf{V}_1 - \mathbf{V}_2 - \mathbf{A}^T\mathbf{\Theta}\mathbf{V}_3 - \sigma^{-1}\mathbf{X}\|_F^2. \quad (24)$$

The solution of (24) can be obtained by solving the following linear system of equations:

$$(\mathbf{I} + \sigma\mathbf{\Theta}^T\mathbf{A}\mathbf{A}^T\mathbf{\Theta})\mathbf{V}_3 = \mathbf{\Theta}^T\mathbf{Y} - \sigma\mathbf{\Theta}^T\mathbf{A}\mathbf{V}_1 - \sigma\mathbf{\Theta}^T\mathbf{A}\mathbf{V}_2 - \mathbf{\Theta}^T\mathbf{A}\mathbf{X}. \quad (25)$$

The minimization of (23) with respect to $\mathbf{V}_1$ and $\mathbf{V}_2$ can be performed as

$$\mathbf{V}_1 = \underset{\mathbf{V}_1}{\operatorname{argmin}} \ p^*(-\mathbf{V}_1) + \frac{\sigma}{2}\|-\mathbf{V}_1 - \mathbf{V}_2 - \mathbf{A}^T\mathbf{\Theta}\mathbf{V}_3 - \sigma^{-1}\mathbf{X}\|_F^2 \quad (26)$$

and

$$\mathbf{V}_2 = \underset{\mathbf{V}_2}{\operatorname{argmin}} \ q^*(-\mathbf{V}_2) + \frac{\sigma}{2}\|-\mathbf{V}_1 - \mathbf{V}_2 - \mathbf{A}^T\mathbf{\Theta}\mathbf{V}_3 - \sigma^{-1}\mathbf{X}\|_F^2. \quad (27)$$

By making use of the Moreau identity, the solutions of (26) and (27) can be obtained in closed forms as

$$\mathbf{V}_1 = \frac{1}{\sigma}\operatorname{Prox}_{\sigma p}(\sigma\mathbf{C}_1) - \mathbf{C}_1 \quad (28)$$

$$\mathbf{V}_2 = \frac{1}{\sigma}\operatorname{Prox}_{\sigma q}(\sigma\mathbf{C}_2) - \mathbf{C}_2 \quad (29)$$

where

$$\mathbf{C}_1 = \mathbf{V}_2 + \mathbf{A}^T\mathbf{\Theta}\mathbf{V}_3 + \sigma^{-1}\mathbf{X} \quad (30)$$

$$\mathbf{C}_2 = \mathbf{V}_1 + \mathbf{A}^T\mathbf{\Theta}\mathbf{V}_3 + \sigma^{-1}\mathbf{X}. \quad (31)$$

$\operatorname{Prox}_p(\cdot)$ denotes the proximal mapping associated with the given closed proper convex function p and is defined by

$$\operatorname{Prox}_p(\mathbf{C}) = \underset{\mathbf{X}}{\operatorname{argmin}} \ p(\mathbf{X}) + \frac{1}{2}\|\mathbf{X} - \mathbf{C}\|_F^2. \quad (32)$$

The process of solving the above proximal mapping is provided in the following. The proximal mapping of σp is written as

$$\operatorname{Prox}_{\sigma p}(\sigma\mathbf{C}_1) = \underset{\mathbf{X}}{\operatorname{argmin}} \ \sigma\alpha\|\mathbf{X}\|_{1,1} + \sigma\beta\|\mathcal{H}_v\mathbf{X}\|_{1,1} + \delta_{\mathbb{R}_+^{m\times n}}(\mathbf{X}) + \frac{1}{2}\|\mathbf{X} - \sigma\mathbf{C}_1\|_F^2. \quad (33)$$

We first define $p_1(\mathbf{X}) = \beta\|\mathcal{H}_v\mathbf{X}\|_{1,1} + \delta_{\mathbb{R}_+^{m\times n}}(\mathbf{X})$ for convenience. From [29, Proposition 1], we can obtain that

$$\begin{aligned}\operatorname{Prox}_{\sigma p}(\sigma\mathbf{C}_1) &= \operatorname{Prox}_{\sigma\alpha\|\cdot\|_{1,1}}(\operatorname{Prox}_{\sigma p_1}(\sigma\mathbf{C}_1)) \\ &= \operatorname{sign}(\operatorname{Prox}_{\sigma p_1}(\sigma\mathbf{C}_1)) \circ \max(|\operatorname{Prox}_{\sigma p_1}(\sigma\mathbf{C}_1)| - \sigma\lambda, 0) \end{aligned} \quad (34)$$

where the proximal mapping of σp_1 can be written as

$$\operatorname{Prox}_{\sigma p_1}(\sigma\mathbf{C}_1) = \underset{\mathbf{X}}{\operatorname{argmin}} \ \delta_{\mathbb{R}_+^{m\times n}}(\mathbf{X}) + \sigma\beta\|\mathcal{H}_v\mathbf{X}\|_{1,1} + \frac{1}{2}\|\mathbf{X} - \sigma\mathbf{C}_1\|_F^2. \quad (35)$$

We further define $p_2(\mathbf{X}) = \beta\|\mathcal{H}_v\mathbf{X}\|_{1,1}$, and then, $\operatorname{Prox}_{\sigma p_1}(\sigma\mathbf{C}_1)$ in (35) can be obtained by

$$\begin{aligned}\operatorname{Prox}_{\sigma p_1}(\sigma\mathbf{C}_1) &= \operatorname{Prox}_{\delta_{\mathbb{R}_+^{m\times n}}(\cdot)}(\operatorname{Prox}_{\sigma p_2}(\sigma\mathbf{C}_1)) \\ &= \Pi_{R_+^{m\times n}}(\operatorname{Prox}_{\sigma p_2}(\sigma\mathbf{C}_1)) \end{aligned} \quad (36)$$

where the proximal mapping of σp_2 can be written as

$$\operatorname{Prox}_{\sigma p_2}(\sigma\mathbf{C}_1) = \underset{\mathbf{X}}{\operatorname{argmin}} \ \sigma\beta\|\mathcal{H}_v\mathbf{X}\|_{1,1} + \frac{1}{2}\|\mathbf{X} - \sigma\mathbf{C}_1\|_F^2. \quad (37)$$

In our proposed framework, we reshape the matrix form of all variables in (37) to vector form and use the 1-D TV denoising algorithm [44] to compute $\operatorname{Prox}_{\sigma p_2}(\sigma\mathbf{C}_1)$.

The proximal mapping of σq is written as

$$\operatorname{Prox}_{\sigma p}(\sigma\mathbf{C}_1) = \underset{\mathbf{X}}{\operatorname{argmin}} \ \sigma\beta\|\mathcal{H}_h\mathbf{X}\|_{1,1} + \frac{1}{2}\|\mathbf{X} - \sigma\mathbf{C}_1\|_F^2. \quad (38)$$

Similar to (37), the 1-D TV denoising algorithm is adopted to compute (38).

Remark 1: The ADMM, which decomposes a difficult problem into a number of simpler subproblems, can also be adopted to solve the optimization problem in (21). More precisely, the ADMM applies AM to the corresponding augmented Lagrangian function with respect to each variable $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$. Nevertheless, the ADMM can only deal with the problem with two blocks with convergence guaranteed. (Note that we regard those separable and independent variables as one block.) As for the three blocks problem in (21), the extended ADMM may not converge. We adopt the sGS-ADMM to solve the problem since the dual of the optimization problem in (21) contains three blocks with two nonsmooth blocks. The sGS-ADMM treats $\mathbf{V}_1$ and $\mathbf{V}_3$ as one block and $\mathbf{V}_2$ as another block and applies AM to the corresponding augmented Lagrangian function with respect to each block $(\mathbf{V}_1, \mathbf{V}_3)$ and $\mathbf{V}_2$. In Algorithm 2, $\mathbf{V}_3$ is updated twice because we use the sGS decomposition to solve the subproblem. The sGS-ADMM we designed in this work, on the other hand, can deal with multiblock convex problems with two nonsmooth blocks with convergence guaranteed.

Next, we establish the global convergence of the sGS-ADMM. We first rewrite problem (21) in the following form:

$$\begin{aligned}&\min_{\mathbf{W},\mathbf{V}_2} \ \psi(\mathbf{W}) + \varphi(\mathbf{V}_2) \\ &\text{s.t.} \ -\mathcal{A}\mathbf{W} - \mathbf{V}_2 = 0 \end{aligned} \quad (39)$$

where

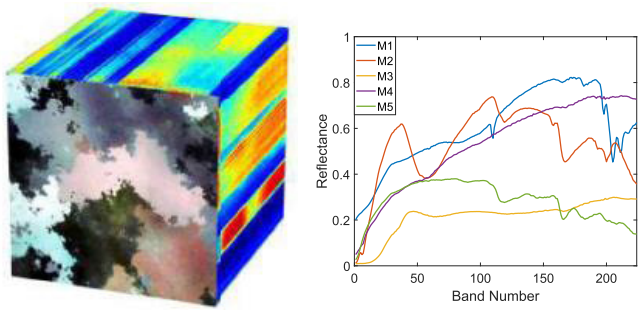
$$\begin{aligned}\psi(\mathbf{W}) &= p^*(-\mathbf{V}_1) + \frac{1}{2}\|\mathbf{V}_3\|_F^2 - \langle \mathbf{V}_3, \mathbf{\Theta}^T\mathbf{Y}\rangle \\ \varphi(\mathbf{V}_2) &= q^*(-\mathbf{V}_2) \end{aligned} \quad (40)$$

and $\mathbf{W} = (\mathbf{V}_1^T, \mathbf{V}_3^T)^T \in \mathbb{R}^{(m+L)\times n}$ and $\mathcal{A} = [\mathbf{I}_{m\times m}, \mathbf{A}^T\mathbf{\Theta}]$. Assume that $(\mathbf{W}, \mathbf{V}_2) \in \mathbb{R}^{(m+L)\times n} \times \mathbb{R}^{m\times n}$ is an optimal solution of (39). If there exists $\mathbf{X} \in \mathbb{R}^{m\times n}$ such that $(\mathbf{W}, \mathbf{V}_2, \mathbf{X})$ satisfies the following Karush-Kuhn-Tucker (KKT) system:

$$\begin{cases} 0 \in \partial\psi(\mathbf{W}) + \mathcal{A}^*\mathbf{X} \\ 0 \in \partial\varphi(\mathbf{V}_2) + \mathbf{X} \\ -\mathcal{A}\mathbf{W} - \mathbf{V}_2 = 0 \end{cases} \quad (41)$$

then $(\mathbf{W}, \mathbf{V}_2, \mathbf{X})$ is a KKT point for the problem (39).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:53:14 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 1. Synthetic data and five endmembers extracted by VCA. (a) Synthetic data. (b) Endmembers.

Theorem 1: Let $\overline{\Omega}$ be the solution set of the KKT system (41) and $\tau \in (0, ((1+\sqrt{5})/2))$. Assume that $\overline{\Omega}$ is nonempty and $\mathbf{S} = (\sigma^{-1}\mathbf{I} + \mathbf{\Theta}^T\mathbf{A}\mathbf{A}^T\mathbf{\Theta}) - \mathbf{\Theta}^T\mathbf{A}[\mathbf{I} + \mathbf{A}^T\mathbf{\Theta}(\sigma^{-1}\mathbf{I} + \mathbf{\Theta}^T\mathbf{A}\mathbf{A}^T\mathbf{\Theta})^{-1}\mathbf{\Theta}^T\mathbf{A}]^{-1}\mathbf{A}^T\mathbf{\Theta}$ is positive definite. Then, there exists a KKT point $\bar{\mathbf{v}} := (\overline{\mathbf{W}}, \overline{\mathbf{V}}_2, \overline{\mathbf{X}}) \in \overline{\Omega}$ such that the sequence $\{(\mathbf{V}_1^j, \mathbf{V}_2^j, \mathbf{V}_3^j, \mathbf{X}^j)\}$ generated by Algorithm 2 converges to $\bar{\mathbf{v}}$.

Proof: See the Appendix.

The computational complexity of Algorithm 2 in each iteration is $O(nL \cdot \max(m, L))$. Note that $L \gg m$, and then, the order of complexity can be simplified to $O(nL^2)$. Similarly, the computational complexity for calculating $\mathbf{\Theta}$ in each iteration is $O(nL^2)$. Thus, the computational complexity of Algorithm 1 in each iteration is $O(jnL^2)$, where j denotes the number of iterations of Algorithm 2.

Remark 2: We should mention that we just give the convergence proof of the sGS-ADMM, while for the AM algorithm, the nonlinear and nonconvex orthogonal constraint makes it quite arduous to prove the convergence in this situation. However, our numerical experiments strongly validate its value and provide evidence of its effectiveness, which motivates future theoretical analysis.

IV. Experiments

In this section, we present the experiments performed on one synthetic dataset and two real datasets to evaluate the proposed approach. The results of the proposed approach have been compared to those obtained with five classical linear unmixing methods, including fully constrained least squares unmixing (FCLSU) [46], partial constrained least squares unmixing (PCLSU) [47], SUnSAL [12], alternating direction sparse and low-rank unmixing (ADSpLRU) [32], and SUnSAL-TV [27], [29], as well as three variability accounting methods, including ELMM [22], ALMM [23], and SULoRA [25]. For a fair comparison, the parameters of all the compared methods are tuned to obtain the best unmixing performance. All experiments are finished by using MATLAB R2017b on a computer with 2.1-GHz Intel Core i7 and 16-GB memory.

To quantitatively assess the performance of the different unmixing methods, we employ the root-mean-square error (RMSE) as the objective metric, which is defined as

$$\text{RMSE} = \frac{1}{n}\sum_{i=1}^{n}\sqrt{\frac{1}{m}\|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2} \tag{42}$$

where $\mathbf{x}_i$ and $\hat{\mathbf{x}}_i$ denote the true abundances and the estimated abundances, respectively. In general, the smaller RMSE values indicate the better estimated unmixing performance. For a reasonable comparison, we average the results of ten runs for all the involved methods. Considering that our unmixing model is a nonconvex optimization problem, the proper initialization of the method would lead to a reasonable and fast solution. The initial value for $\mathbf{X}^0$ is set as the output of PCLSU. The initial value for $\mathbf{\Theta}^0$ is set to $[\mathbf{I}_{r\times r}; \mathbf{0}_{(L-r)\times r}]$. The initial values for $\mathbf{V}_1^0$, $\mathbf{V}_2^0$, and $\mathbf{V}_3^0$ are set equal to $\mathbf{0}$.

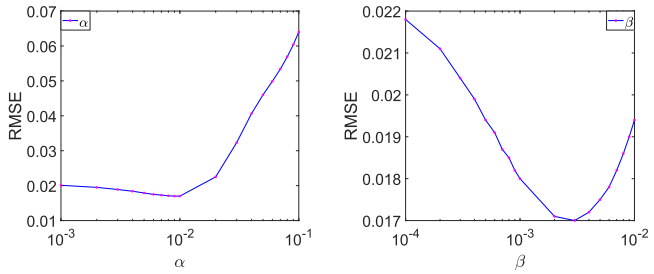


Fig. 2. Parameter sensitivity analysis for two regularization parameters (α and β) on the synthetic dataset.

A. Synthetic Dataset Experiments

1) Data Description: The synthetic dataset contains 200 × 200 pixels with 224 spectral bands that are uniformly distributed from 0.4 to 2.5 μm. Five endmembers corresponding to the signatures of the minerals are randomly selected from the USGS digital library. Fig. 1 shows the synthetic data and five endmembers extracted by VCA. For each pixel, the abundance maps are generated following the Gaussian field and satisfy the ANC and ASC. Note that each pixel includes spectral variability due to the complex noise and endmember-dependent scaling factors. Specifically, the given reference endmembers first multiply randomly generated scaling factors ranging in [0.75, 1.25], and a 25-dB white Gaussian noise is added to these scaled endmembers. Then, the LMM is performed by using the generated abundance maps. Finally, a 25-dB white Gaussian noise is added to the generated pixels again. This synthetic dataset with such spectral variability can give us a proper scenario to validate the proposed methods. For more details on this dataset, refer to [22].

2) Experimental Setup: The endmember matrix of all the involved methods is constructed via VCA [8], while Hysime [42] is adopted to estimate the number of endmembers first. Then, the endmembers can be identified by calculating the spectral angle of the estimated and reference endmembers. To fairly assess the unmixing performance, the optimal parameters for all the involved methods are set and recorded as follows. The sparsity regularization of the SUnSAL is parameterized by $2e^{-3}$. For the ADSpLRU, the sparsity and low-rank regularizations are parameterized by $\{5e^{-2}, 1e^{-2}\}$. The sparsity and TV regularizations for the SUnSAL-TV are parameterized by $\{1e^{-2}, 3e^{-3}\}$. The regularization parameter for the ELMM is set as $5e^{-1}$. The parameters for the ALMM are set as $\{2e^{-3}, 2e^{-3}, 5e^{-3}, 5e^{-3}\}$, and the SULoRA's parameters are $\{1e^{-1}, 1e^{-2}, 8e^{-3}\}$. The parameters of the proposed OSU method are set as $\{1e^{-2}, 3e^{-3}\}$.

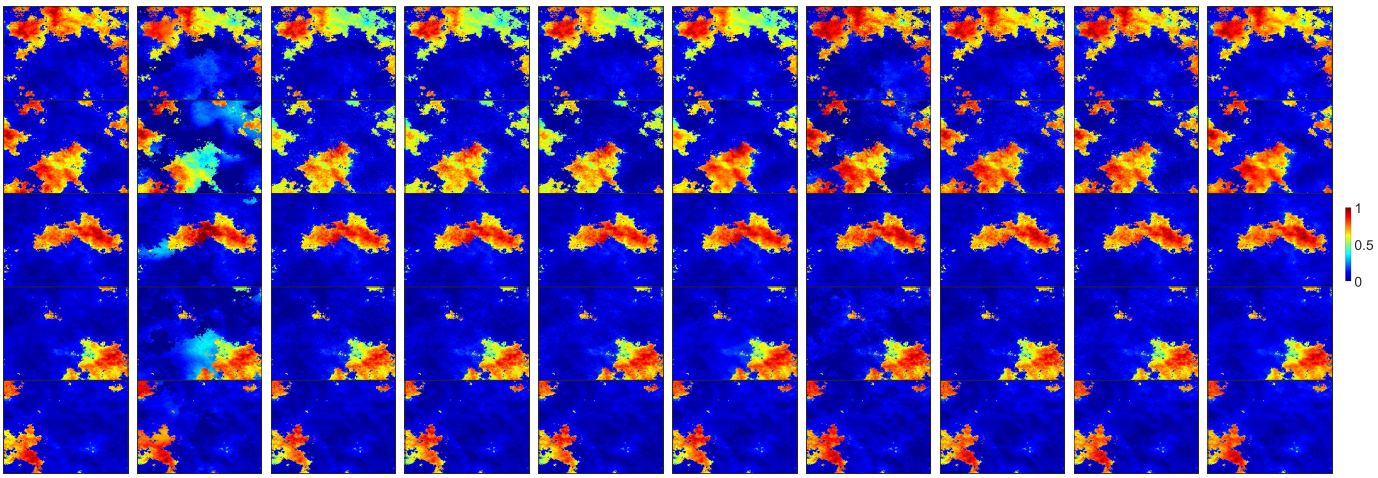
3) Parameters Sensitivity Analysis: As the performance of the proposed OSU method is sensitive to the parameters α and β, it is essential to find a suitable range to guide the selection of the parameters. For this purpose, a grid search strategy

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:53:14 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE RESULTS OF THE NUMERICAL EXPERIMENTS ON THE SYNTHETIC DATASET. THE BEST ONE IS MARKED IN BOLD

Metric	Ideal Value	Method								
		FCLSU	PCLSU	SUnSAL	ADSpLRU	SUnSAL-TV	ELMM	ALMM	SULoRA	OSU
RMSE	0	0.0588	0.0401	0.0392	0.0387	0.0328	0.0327	0.0214	0.0220	**0.0170**
Time (Sec.)	-	0.51	0.11	**0.06**	14.12	4.93	146.41	74.23	3.19	3.01



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 3. Reference abundance maps and the estimated abundance maps obtained by using all the tested methods for the synthetic dataset (each row corresponds to one endmember). (a) Reference abundance maps. (b) FCLSU. (c) PCLSU. (d) SUnSAL. (e) ADSpLRU. (f) SUnSAL-TV. (g) ELMM. (h) ALMM. (i) SULoRA. (j) OSU.

is adopted throughout our experiments. More specifically, the optimal value of each parameter is evaluated, while the other one is fixed until the lowest RMSE values are reached. Fig. 2 plots the RMSE values as a function of the two regularization parameters on the synthetic dataset. One can observe that the proposed method achieves high RMSE values when either α is too large or β is too small, and the RMSE values decrease as the parameter α decreases or β increases until the lowest RMSE value is achieved. This phenomenon demonstrates the positive influence of the proper selection of the two parameters on the unmixing performance. The optimal parameters $\alpha = 1e^{-2}$ and $\beta = 3e^{-3}$ can be found with approximately convex curves.

4) Results and Analysis: Table I lists the RMSE values and the running time for each unmixing method on the synthetic dataset. From Table I, we can observe that the classical linear methods, including FCLSU, PCLSU, SUnSAL, ADSpLRU, and SUnSAL-TV, yield poor results due to the lack of characterization for spectral variability, while for the methods considering the spectral variability, our proposed OSU method achieves the smallest RMSE values. In terms of running time, the FCLSU, PCLSU, and SUnSAL are fast due to the simple implementation, followed by our OSU method which is slightly slower. Also, the rest of the methods take more time, especially for the ELMM and ALMM.

A visual representation of the reference abundance maps and the estimated abundance maps obtained by using all the tested methods for the synthetic dataset is shown in Fig. 3. One can clearly observe that the FCLSU yields the poorest performance since the ASC in FCLSU fully absorbs the spectral variabilities. The PCLSU obtains more accurate abundance maps than FCLSU since it drops the ASC and estimates the abundances in a cone rather than the simplex. The SUnSAL constrains the single sparsity on the abundance, providing similar behavior as the PCLSU since the presence of spectral variability. The ADSpLRU and SUnSAL-TV slightly improve the performance by exploiting the spatial information. All the above five unmixing methods actually do not eliminate the spectral variabilities but are rather absorbed by the abundances and thus obtain an inaccurate solution. Taking the remaining methods, by and large, those of modeling the spectral variabilities provide more accurate abundance maps than those without considering ones. The ELMM is able to take account into the scaling factors, but the estimated abundance maps are noisy. The ALMM obviously improves the performance by considering the scaling factors and other variabilities. With the sparsity prior and low-rank attribute, the SULoRA also tends to hold fine unmixing performance. As expected, the performance of the OSU method outperforms all the compared methods, revealing the superiority of conducting the unmixing in the robust orthogonal subspace.

We also show the difference maps between the true and estimated abundance maps obtained by using all the tested methods in Fig. 4. Visually, Fig. 4(a)–(e) shows many noisy regions for each endmember due to the lack of incorporation of spectral variability, among which Fig. 4(a) is the most serious, while those methods considering spectral variability show fewer noise regions as shown in Fig. 4(f)–(i), among which Fig. 4(i) is the closest to zero. These results reveal the superiority of the proposed method to deal with spectral variability.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:53:14 UTC from IEEE Xplore. Restrictions apply.

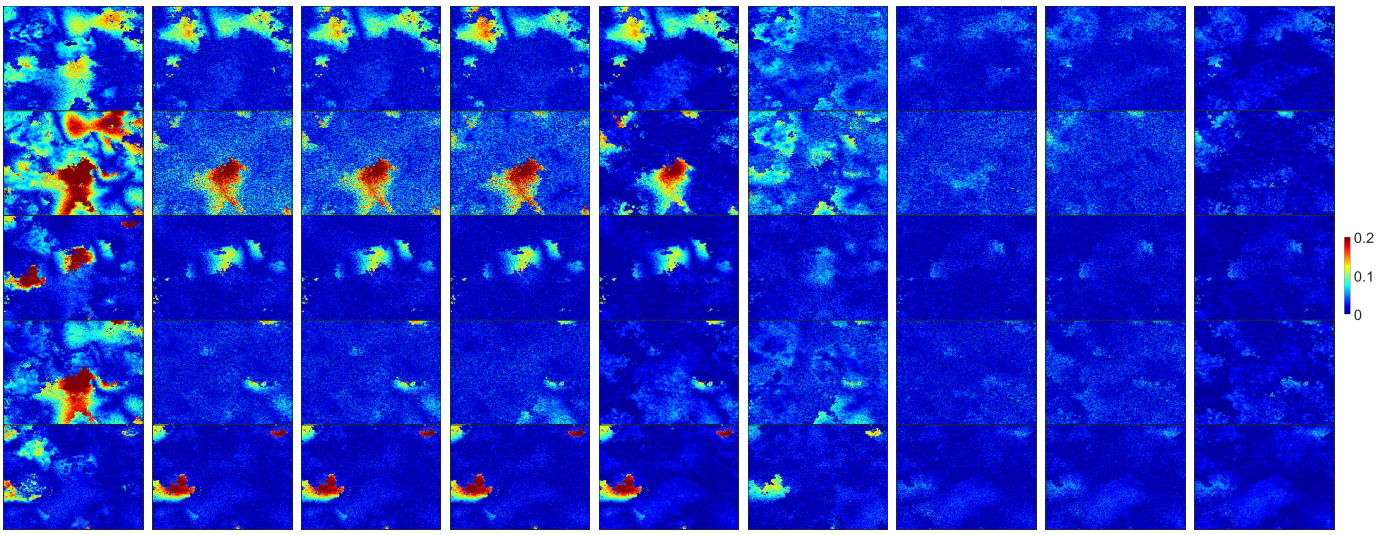


Fig. 4. Difference maps between the true and estimated abundance maps for the synthetic dataset. (a) FCLSU. (b) PCLSU. (c) SUnSAL. (d) ADSpLRU. (e) SUnSAL-TV. (f) ELMM. (g) ALMM. (h) SULoRA. (i) OSU.

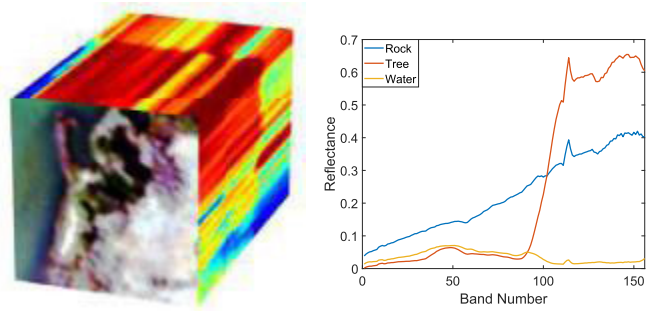


Fig. 5. Samson data and three endmembers extracted by VCA. (a) Samson data. (b) Endmembers.

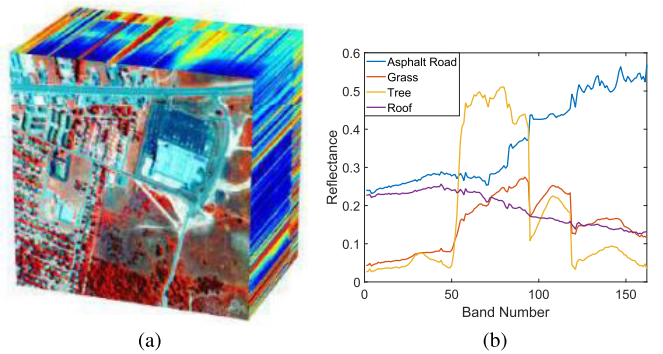


Fig. 6. Urban data and four endmembers extracted by VCA. (a) Urban data. (b) Endmembers.

B. Samson Dataset Experiments

1) Data Description: The first real dataset we used is the publicly available Samson dataset, which is quite prevalent for hyperspectral analysis. We selected a 95 × 95 pixels data subset to evaluate the unmixing performance between the proposed method and the compared methods, as shown in Fig. 5(a). This dataset contains 156 spectral bands ranging from 401 nm to 889 nm with a 3.13-nm spectral resolution. Prior to the analysis, these data are not degraded by the badly noised channels or blank channels.

2) Experimental Setup: Similar to the synthetic dataset, the Hysime is adopted first to estimate the number of the endmember, and then, the endmembers are extracted from the data. In addition, the endmembers identification step is conducted by comparison with the reference endmembers.[2] Three endmembers used in the unmixing process, including the rock, tree, and water, are shown in Fig. 5(b). The optimal parameter setting for all the methods is recorded as follows. For the SUnSAL, the sparsity regularization is parameterized by $6e^{-3}$. The sparsity and low-rank regularizations for the ADSpLRU are parameterized by $\{1e^{-1}, 1\}$. The sparsity and TV regularizations for the SUnSAL-TV are parameterized by $\{5e^{-3}, 1e^{-2}\}$. The regularization parameter for the ELMM is set as $5e^{-1}$. The parameters for the ALMM are set as $\{2e^{-4}, 2e^{-2}, 4e^{-3}, 1e^{-3}\}$, and the SULoRA's parameters are $\{1, 1e^{-2}, 2e^{-2}\}$. The parameters of the OSU method are set as $\{6e^{-2}, 1e^{-2}\}$.

3) Results and Analysis: Table II lists the RMSE values and the running time for each method on the Samson dataset. Overall, the classic linear unmixing methods achieve high RMSE values as they fail to character the spectral variability. The ELMM, ALMM, and SULoRA can explain the spectral variabilities partly and achieves relatively low RMSE values, while for the proposed OSU method, the lowest RMSE value is achieved. Furthermore, the proposed OSU method also exhibits great competitiveness in terms of running time. For visual comparison, Fig. 7 shows the reference abundances maps and the estimated abundance maps obtained by using all the tested methods for endmembers: rock, tree, and water. At first glance, all the methods seem to produce similar distributions of the endmembers of the interest. In more detail, the FCLSU performs a poor visualization on the endmember of water, while the PCLSU, SUnSAL, ADSpLRU, and SUnSAL-TV provide a more realistic distribution of the water, but a poor visualization of the rock and tree. As expected, the methods modeling the spectral variability produce more realistic abundance maps. However, one can observe that the

[2]Available online: https://rslab.ut.ac.ir/data

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:53:14 UTC from IEEE Xplore. Restrictions apply.

TABLE II

QUANTITATIVE RESULTS OF THE NUMERICAL EXPERIMENTS ON THE SAMSON DATASET. THE BEST ONE IS MARKED IN BOLD

Metric	Ideal Value	Method								
		FCLSU	PCLSU	SUnSAL	ADSpLRU	SUnSAL-TV	ELMM	ALMM	SULoRA	OSU
RMSE	0	0.1505	0.1410	0.1344	0.1343	0.1338	0.1184	0.0995	0.0919	**0.0712**
Time (Sec.)	-	0.05	0.03	**0.02**	15.74	0.72	59.62	28.75	0.71	0.51

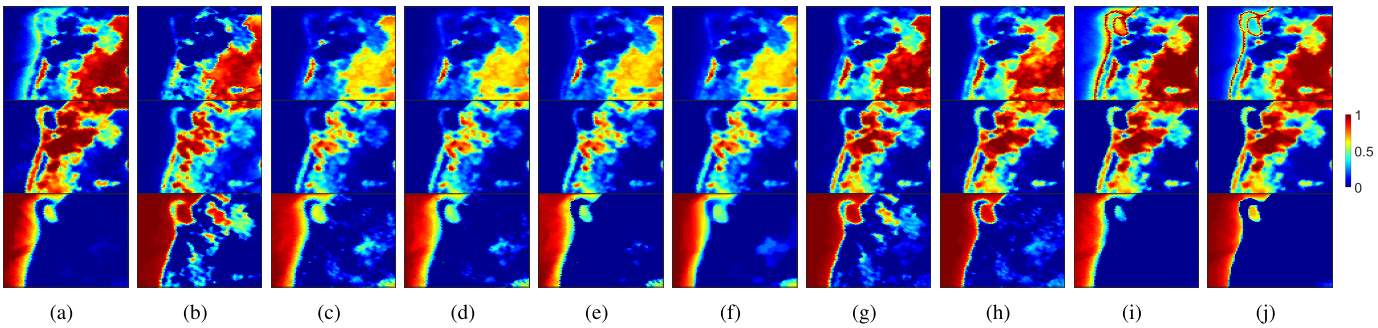


Fig. 7. Reference abundance maps and the estimated abundance maps obtained by using all the tested methods for the Samson dataset (from top to bottom: rock, tree, and water). (a) Reference abundance maps. (b) FCLSU. (c) PCLSU. (d) SUnSAL. (e) ADSpLRU. (f) SUnSAL-TV. (g) ELMM. (h) ALMM. (i) SULoRA. (j) OSU.

TABLE III

QUANTITATIVE RESULTS OF THE NUMERICAL EXPERIMENTS ON THE URBAN DATASET. THE BEST ONE IS MARKED IN BOLD

Metric	Ideal Value	Method								
		FCLSU	PCLSU	SUnSAL	ADSpLRU	SUnSAL-TV	ELMM	ALMM	SULoRA	OSU
RMSE	0	0.2583	0.1958	0.1935	0.1964	0.1785	0.1410	0.1366	0.1076	**0.0979**
Time (Sec.)	-	0.90	0.88	**0.73**	89.93	7.42	494.66	255.89	6.23	4.81

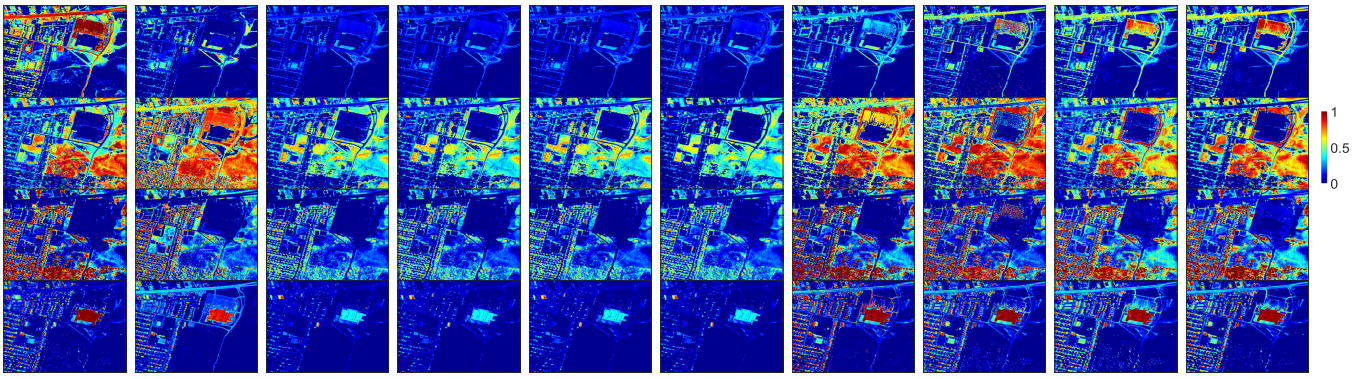


Fig. 8. Reference abundance maps and the estimated abundance maps obtained by using all the tested methods for the Urban dataset (from top to bottom: asphalt, grass, tree, and roof). (a) Reference abundance maps. (b) FCLSU. (c) PCLSU. (d) SUnSAL. (e) ADSpLRU. (f) SUnSAL-TV. (g) ELMM. (h) ALMM. (i) SULoRA. (j) OSU.

ELMM and ALMM inaccurately estimated the endmember of water, whereas the estimated abundance maps produced by the proposed OSU method are much closer to the reference abundance maps than SULoRA. These results further confirm the superiority of the proposed OSU method to not only the classic unmixing methods but also the variability accounting methods.

C. Urban Dataset Experiments

1) Data Description: The second real dataset we used is the Urban data captured by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor over an area located as Copperas Cove, TX, USA, in October 1995. These data contain 307 × 307 pixels, measured in 210 spectral bands ranging from 400 to 2500 nm with a 10-nm spectral resolution, as shown in Fig. 6(a). Prior to the analysis, we remove bands 1–4, 76, 87, 101–111, 136–153, and 198–210 since the interference of water absorption and noise, leaving a total of 162 effective spectral bands.

2) Experimental Setup: Likewise, the endmembers are extracted from the data through the Hysime and VCA and then identified by using the spectral angle of the estimated and reference endmembers. We record the optimal parameters for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:53:14 UTC from IEEE Xplore. Restrictions apply.

all the methods as follows. The parameters for the compared methods on these data are $6e^{-3}$ for SUnSAL, $\{1e^{-2}, 1e^{-2}\}$ for ADSpLRU, $\{5e^{-2}, 5e^{-3}\}$ for SUnSAL-TV, $5e^{-1}$ for ELMM, $\{5e^{-2}, 5e^{-2}, 1e^{-2}, 1e^{-2}\}$ for ALMM, and $\{1e^{-1}, 1e^{-2}, 5e^{-3}\}$ for SULoRA. The parameters in the OSU method are set as $\{5e^{-2}, 5e^{-3}\}$.

3) Results and Analysis: Table III lists the RMSE values and the running time for each method on the Urban dataset. The reference abundance maps and the estimated abundance maps obtained by using all the tested methods for endmembers: asphalt, grass, tree, and roof are also shown in Fig. 8. According to the quantitative and qualitative visual results, we analyze the performance of the different methods as follows. Similarly, the classic unmixing methods yield relatively high RMSE values, whereas the proposed OSU method accounts for the spectral variability in a more generalized way resulting in a more desirable result than the other variability accounting methods. Visually, for the asphalt, the classic unmixing methods fail to capture it, while both the ELMM and ALMM improperly unmix it as the ingredient of the grass. Both SULoRA and OSU accurately estimated the asphalt, whereas our method produces more clear abundance map. The estimated abundance maps of other endmembers in Fig. 8 give consistent results. These results reveal the potential of the proposed OSU method to handle spectral variability in a generalized way by using the orthogonal subspace.

V. Conclusion

In this article, we exhibited experimental evidence of orthogonal subspace learning in mitigating the effects of spectral variability. The proposed OSU model provides a new manner to conduct unmixing in the orthogonal subspace rather than the original space. One major enrichment of the proposed model is that it does not assume any special fabric to depict the endmember signature. Extensive numerical experiments on one synthetic and two real hyperspectral data demonstrated the effectiveness and superiority of the proposed method over classical linear unmixing methods or variability accounting approaches. Future work will focus on adapting the orthogonal subspace-based framework to be applicable to the tensor-based unmixing methods. Moreover, proving the convergence of the proposed algorithm is also a promising topic.

Appendix
Proof of Theorem 1

Proof: From [48, Th. 12.17], we know that the subdifferential mappings of the closed proper convex functions are maximally monotone, that is, there exist two self-adjoint and positive semidefinite linear operators Σ_ψ and Σ_φ such that for all $\mathbf{y}, \mathbf{y}' \in \mathrm{dom}(\psi)$, $\xi \in \partial\psi(\mathbf{y})$ and $\xi' \in \partial\psi(\mathbf{y}')$

$$\psi(\mathbf{y}) \geq \psi(\mathbf{y}') + \langle \xi', \mathbf{y} - \mathbf{y}' \rangle + \frac{1}{2}\|\mathbf{y} - \mathbf{y}'\|^2_{\Sigma_\psi} \tag{43}$$

$$\langle \xi - \xi', \mathbf{y} - \mathbf{y}' \rangle \geq \|\mathbf{y} - \mathbf{y}'\|^2_{\Sigma_\psi} \tag{44}$$

and for all $\mathbf{z}, \mathbf{z}' \in \mathrm{dom}(\varphi)$, $\zeta \in \partial\varphi(\mathbf{z})$ and $\zeta' \in \partial\varphi(\mathbf{z}')$

$$\varphi(\mathbf{z}) \geq \varphi(\mathbf{z}') + \langle \zeta', \mathbf{z} - \mathbf{z}' \rangle + \frac{1}{2}\|\mathbf{z} - \mathbf{z}'\|^2_{\Sigma_\varphi} \tag{45}$$

$$\langle \zeta - \zeta', \mathbf{z} - \mathbf{z}' \rangle \geq \|\mathbf{z} - \mathbf{z}'\|^2_{\Sigma_\varphi}. \tag{46}$$

From [45], we know that performing a one-cycle sGS method is equivalent to solving the associated convex quadratic programming problem plus an additional semiproximal term $(1/2)\|\mathbf{W} - \mathbf{W}^k\|^2_{\mathcal{T}}$, where $\mathcal{T}$ is a symmetric positive semidefinite operator related to the sGS decomposition and $\mathbf{W}^k$ is the previous iterate. According to [49, Th. B.1], in order to prove the convergence of the proposed framework, we need to establish the following two conditions hold:

$$\Sigma_\varphi + \sigma\mathbf{I} \succ 0 \tag{47}$$

$$\Sigma_\psi + \sigma\mathcal{A}^*\mathcal{A} + \mathcal{T} \succ 0 \tag{48}$$

where

$$\Sigma_\psi := \begin{pmatrix} 0 & 0 \\ 0 & \mathbf{I} \end{pmatrix} \tag{49}$$

$$\begin{aligned} \mathcal{T} &:= \begin{pmatrix} 0 & \sigma\mathbf{A}^T\boldsymbol{\Theta} \\ 0 & 0 \end{pmatrix} \\ &\quad \begin{pmatrix} \sigma^{-1}\mathbf{I} & 0 \\ 0 & (\mathbf{I} + \sigma\boldsymbol{\Theta}^T\mathbf{A}\mathbf{A}^T\boldsymbol{\Theta})^{-1} \end{pmatrix} \begin{pmatrix} 0 & 0 \\ \sigma\boldsymbol{\Theta}^T\mathbf{A} & 0 \end{pmatrix} \\ &= \begin{pmatrix} \sigma\mathbf{A}^T\boldsymbol{\Theta}(\sigma^{-1}\mathbf{I} + \boldsymbol{\Theta}^T\mathbf{A}\mathbf{A}^T\boldsymbol{\Theta})^{-1}\boldsymbol{\Theta}^T\mathbf{A} & 0 \\ 0 & 0 \end{pmatrix}. \end{aligned} \tag{50}$$

It is obvious that (47) holds directly. By simple algebra, (48) is equivalent to that

$$\begin{pmatrix} \mathbf{I} + \mathbf{A}^T\boldsymbol{\Theta}(\sigma^{-1}\mathbf{I} + \boldsymbol{\Theta}^T\mathbf{A}\mathbf{A}^T\boldsymbol{\Theta})^{-1}\boldsymbol{\Theta}^T\mathbf{A} & \mathbf{A}^T\boldsymbol{\Theta} \\ \boldsymbol{\Theta}^T\mathbf{A} & \sigma^{-1}\mathbf{I} + \boldsymbol{\Theta}^T\mathbf{A}\mathbf{A}^T\boldsymbol{\Theta} \end{pmatrix} \tag{51}$$

is positive definite. From the Schur complement condition [50], $\mathbf{S}$ is required to be positive definite to ensure that (51) is positive definite.

References

[1] A. F. H. Goetz, "Three decades of hyperspectral remote sensing of the Earth: A personal view," *Remote Sens. Environ.*, vol. 113, pp. S5–S16, Sep. 2009.

[2] D. Hong et al., "Interpretable hyperspectral artificial intelligence: When non-convex modeling meets hyperspectral remote sensing," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 2, pp. 52–87, Jun. 2021.

[3] R. A. Borsoi et al., "Spectral variability in hyperspectral data unmixing: A comprehensive review," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 223–270, Apr. 2021.

[4] X. Wu, D. Hong, and J. Chanussot, "UIU-Net: U-Net in U-Net for infrared small object detection," *IEEE Trans. Image Process.*, vol. 32, pp. 364–376, 2023.

[5] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[6] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[7] J. M. Bioucas-Dias et al., "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[8] J. M. P. Nascimento and J. M. Bioucas-Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

[9] M. E. Winter, "N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data," *Proc. SPIE*, vol. 3753, pp. 266–275, Oct. 1999.

[10] J. Li and J. M. Bioucas-Dias, "Minimum volume simplex analysis: A fast algorithm to unmix hyperspectral data," in *Proc. IEEE IGARSS*, vol. 3, pp. 250–253, 2008.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:53:14 UTC from IEEE Xplore. Restrictions apply.

[11] D. A. Roberts, M. Gardner, R. Church, S. Ustin, G. Scheer, and R. O. Green, "Mapping chaparral in the Santa Monica Mountains using multiple endmember spectral mixture models," *Remote Sens. Environ.*, vol. 65, no. 3, pp. 267–279, Sep. 1998.
[12] M.-D. Iordache, J. Bioucas-Dias, and A. Plaza, "Sparse unmixing of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 6, pp. 2014–2039, Jun. 2011.
[13] J. Yao, D. Meng, Q. Zhao, W. Cao, and Z. Xu, "Nonconvex-sparsity and nonlocal-smoothness-based blind hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 2991–3006, Jun. 2019.
[14] L. Ren, Z. Ma, F. Bovolo, and L. Bruzzone, "A nonconvex framework for sparse unmixing incorporating the group structure of the spectral library," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5506719.
[15] L. Ren, Z. Ma, and F. Bovolo, "A novel dual-alternating direction method of multipliers for spectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 3, pp. 528–532, Mar. 2021.
[16] W. He, H. Zhang, and L. Zhang, "Sparsity-regularized robust nonnegative matrix factorization for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 9, pp. 4267–4279, Sep. 2016.
[17] L. Drumetz, T. R. Meyer, J. Chanussot, A. L. Bertozzi, and C. Jutten, "Hyperspectral image unmixing with endmember bundles and group sparsity inducing mixed norms," *IEEE Trans. Image Process.*, vol. 28, no. 7, pp. 3435–3450, Jul. 2019.
[18] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "CoSpace: Common subspace learning from hyperspectral-multispectral correspondences," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4349–4359, Jul. 2019.
[19] X. Du, A. Zare, P. Gader, and D. Dranishnikov, "Spatial and spectral unmixing using the beta compositional model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 1994–2003, Jun. 2014.
[20] O. Eches, N. Dobigeon, C. Mailhes, and J.-Y. Tourneret, "Bayesian estimation of linear mixtures using the normal compositional model. Application to hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 19, no. 6, pp. 1403–1413, Jun. 2010.
[21] P.-A. Thouvenin, N. Dobigeon, and J.-Y. Tourneret, "Hyperspectral unmixing with spectral variability using a perturbed linear mixing model," *IEEE Trans. Signal Process.*, vol. 64, no. 2, pp. 525–538, Jan. 2015.
[22] L. Drumetz, M.-A. Veganzones, S. Henrot, R. Phlypo, J. Chanussot, and C. Jutten, "Blind hyperspectral unmixing using an extended linear mixing model to address spectral variability," *IEEE Trans. Image Process.*, vol. 25, no. 8, pp. 3890–3905, Aug. 2016.
[23] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.
[24] S. Henrot, J. Chanussot, and C. Jutten, "Dynamical spectral unmixing of multitemporal hyperspectral images," *IEEE Trans. Image Process.*, vol. 25, no. 7, pp. 3219–3232, Jul. 2016.
[25] D. Hong and X. Zhu, "SULoRA: Subspace unmixing with low-rank attribute embedding for hyperspectral data analysis," *IEEE J. Sel. Topics Signal Process.*, vol. 12, no. 6, pp. 1351–1363, Dec. 2018.
[26] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Collaborative sparse regression for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 341–354, Jan. 2014.
[27] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Total variation spatial regularization for sparse hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 11, pp. 4484–4502, Nov. 2011.
[28] L. Ren, Z. Ma, F. Bovolo, J. Hu, and L. Bruzzone, "Hyperspectral unmixing using weighted sparse regression with total variation regularization," *Int. J. Remote. Sens.*, vol. 43, nos. 15–16, pp. 1–28, 2022.
[29] L. Ren, C. Wang, P. Tang, and Z. Ma, "A dual symmetric Gauss–Seidel alternating direction method of multipliers for hyperspectral sparse unmixing," *Numer. Algorithms*, vol. 87, no. 2, pp. 719–754, Jun. 2021.
[30] W. He, H. Zhang, and L. Zhang, "Total variation regularized reweighted sparse nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3909–3921, Jul. 2017.
[31] X.-L. Zhao, F. Wang, T.-Z. Huang, M. K. Ng, and R. J. Plemmons, "Deblurring and sparse unmixing for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 7, pp. 4045–4058, Jul. 2013.
[32] P. V. Giampouras, K. E. Themelis, A. A. Rontogiannis, and K. D. Koutroumbas, "Simultaneously sparse and low-rank abundance matrix estimation for hyperspectral image unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4775–4789, Aug. 2016.
[33] J. Huang, T.-Z. Huang, L.-J. Deng, and X.-L. Zhao, "Joint-sparse-blocks and low-rank representation for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2419–2438, Apr. 2019.
[34] L. Ren, D. Hong, X. Sun, L. Gao, and M. Huang, "Hyperspectral sparse unmixing via firm thresholding mapping," in *Proc. IEEE WHISPERS*, Sep. 2022, pp. 1–5.
[35] L. Ren, D. Hong, L. Gao, X. Sun, M. Huang, and J. Chanussot, "Hyperspectral sparse unmixing via nonconvex shrinkage penalties," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500415.
[36] J. Yao, D. Hong, L. Xu, D. Meng, J. Chanussot, and Z. Xu, "Sparsity-enhanced convolutional decomposition: A novel tensor-based paradigm for blind hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5505014.
[37] D. Hong, N. Yokoya, J. Chanussot, J. Xu, and X. X. Zhu, "Learning to propagate labels on graphs: An iterative multitask regression framework for semi-supervised hyperspectral dimensionality reduction," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 35–49, Dec. 2019.
[38] M. Wang, Q. Wang, J. Chanussot, and D. Hong, "l_0-l_1 hybrid total variation regularization and its applications on hyperspectral image mixed noise removal and compressed sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7695–7710, Sep. 2021.
[39] M. Wang, Q. Wang, and J. Chanussot, "Tensor low-rank constraint and ℓ_0 total variation for hyperspectral image mixed noise removal," *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 3, pp. 718–733, Feb. 2021.
[40] D. Hong, N. Yokoya, N. Ge, J. Chanussot, and X. X. Zhu, "Learnable manifold alignment (LeMA): A semi-supervised cross-modality learning framework for land cover and land use classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 147, pp. 193–205, Jan. 2019.
[41] J. Lin, T.-Z. Huang, X.-L. Zhao, T.-X. Jiang, and L. Zhuang, "A tensor subspace representation-based method for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7739–7757, Sep. 2021.
[42] J. M. Bioucas-Dias and J. M. P. Nascimento, "Hyperspectral subspace identification," *IEEE Trans. Geosci. Remote Sensing*, vol. 46, no. 8, pp. 2435–2445, Aug. 2008.
[43] A. Beck and M. Teboulle, "Fast gradient-based algorithms for constrained total variation image denoising and deblurring problems," *IEEE Trans. Image Process.*, vol. 18, no. 11, pp. 2419–2434, Nov. 2009.
[44] L. Condat, "A direct algorithm for 1-D total variation denoising," *IEEE Signal Process. Lett.*, vol. 20, no. 11, pp. 1054–1057, Nov. 2013.
[45] X. Li, D. Sun, and K.-C. Toh, "A block symmetric Gauss–Seidel decomposition theorem for convex composite quadratic programming and its applications," *Math. Program.*, vol. 175, nos. 1–2, pp. 395–418, May 2019.
[46] D. C. Heinz and C.-I. Chang, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.
[47] R. Heylen, D. Burazerovic, and P. Scheunders, "Fully constrained least squares spectral unmixing by simplex projection," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4112–4122, Nov. 2011.
[48] R. T. Rockafellar and R. J.-B. Wets, *Variational Analysis*, vol. 317. Berlin, Germany: Springer, 2009.
[49] M. Fazel, T. Pong, D. Sun, and P. Tseng, "Hankel matrix rank minimization with applications to system identification and realization," *SIAM J. Matrix Anal. Appl.*, vol. 34, no. 3, pp. 946–977, 2013.
[50] R. A. Horn and C. R. Johnson, *Matrix Analysis*. Cambridge, U.K.: Cambridge Univ. Press, 2012.

Longfei Ren received the Ph.D. degree from the School of Information Science and Technology, Southwest Jiaotong University, Chengdu, China, in 2021.

From 2019 to 2020, he was a Visiting Ph.D. Student with the Remote Sensing Laboratory, Department of Information and Communication Technologies, University of Trento, Trento, Italy, and the Remote Sensing for Digital Earth Unit, Fondazione Bruno Kessler, Trento. He is currently working with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include hyperspectral unmixing, optimization, and image processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:53:14 UTC from IEEE Xplore. Restrictions apply.

Danfeng Hong (Senior Member, IEEE) received the M.Sc. degree (summa cum laude) in computer vision from the College of Information Engineering, Qingdao University, Qingdao, China, in 2015, and the Dr.Ing degree (summa cum laude) from the Signal Processing in Earth Observation (SiPEO), Technical University of Munich (TUM), Munich, Germany, in 2019.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences (CAS), Beijing, China. Before joining CAS, he has been a Research Scientist and led a Spectral Vision Working Group at the Remote Sensing Technology Institute (IMF), German Aerospace Center (DLR), Oberpfaffenhofen, Germany. He was an Adjunct Scientist at GIPSA-Lab, Grenoble INP, CNRS, University of Grenoble Alpes, Grenoble, France. His research interests include signal/image processing, hyperspectral remote sensing, machine/deep learning, artificial intelligence, and their applications in Earth vision.

Dr. Hong was a recipient of the Best Reviewer Award of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) in 2021 and 2022, and the Best Reviewer Award of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING in 2022, the Jose Bioucas Dias Award for recognizing the outstanding paper at WHISPERS in 2021, the Remote Sensing Young Investigator Award in 2022, the IEEE GRSS Early Career Award in 2022, and a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters) in 2022. He is an Associate Editor of the TGRS, an Editorial Board Member of *Remote Sensing*, and an Editorial Advisory Board Member of *ISPRS Journal of Photogrammetry and Remote Sensing*

Lianru Gao (Senior Member, IEEE) received the B.S. degree in civil engineering from Tsinghua University, Beijing, China, in 2002, and the Ph.D. degree in cartography and geographic information system from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, in 2007.

He has been a Visiting Scholar with the University of Extremadura, Cáceres, Spain, in 2014, and Mississippi State University (MSU), Starkville, MS, USA, in 2016. He is currently a Professor with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, CAS. In the last ten years, he was the Principal Investigator of ten scientific research projects at national and ministerial levels, including projects by the National Natural Science Foundation of China from 2016 to 2019, 2018 to 2020, and 2022 to 2025, and the Key Research Program of the CAS from 2013 to 2015. He has published more than 180 peer-reviewed articles, where more than 100 journal articles are included in the Science Citation Index (SCI). He was a coauthor of three academic books, including *Hyperspectral Image Information Extraction*. He obtained 29 national invention patents in China. His research focuses on hyperspectral image processing and information extraction.

Dr. Gao was awarded the Outstanding Science and Technology Achievement Prize of the CAS in 2016 and was supported by the China National Science Fund for Excellent Young Scholars in 2017. He received the Second Prize of the State Scientific and Technological Progress Award in 2018, the recognition of the Best Reviewers of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING in 2015, and the Best Reviewers of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING in 2017.

Xu Sun (Member, IEEE) received the B.S. degree in mathematics and application mathematics from Tsinghua University, Beijing, China, in 2006, and the Ph.D. degree in cartography and geographical information system from the Graduate University of Chinese Academy of Sciences, Beijing, in 2011.

He is currently an Associate Researcher with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include hyperspectral image processing, artificial intelligence algorithm, and high-resolution remote sensing image information mining.

Min Huang received the B.S. degree from the University of Science and Technology of China, Hefei, China, in 1999, and the Ph.D. degree in optical engineering from the Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences (CAS), Xi'an, China, in 2009.

He is currently a Professor with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, CAS, and also with the School of Optoelectronics, University of Chinese Academy of Science. In the last ten years, he was the Principal Investigator of ten scientific research projects at national and ministerial levels, including projects by the Key Research Program of CAS Support Technology and the National High-Tech Research and Development Program. He has published more than 30 peer-reviewed articles and obtained more than 20 national invention patents in China. His research focuses on imaging spectrometry and computational optical imaging.

Dr. Huang received the Second Prize of the State Scientific and Technological Progress Award in 2010.

Jocelyn Chanussot (Fellow, IEEE) received the M.Sc. degree in electrical engineering from the Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Universite de Savoie, Annecy, France, in 1998.

Since 1999, he has been with Grenoble INP, where he is currently a Professor of signal and image processing. He has been a Visiting Scholar with Stanford University, Stanford, CA, USA; the KTH Royal Institute of Technology, Stockholm, Sweden; and the National University of Singapore (NUS), Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavík, Iceland. From 2015 to 2017, he was a Visiting Professor with the University of California at Los Angeles (UCLA), Los Angeles, CA, USA. He holds the AXA Chair in remote sensing and is also an Adjunct Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include image analysis, hyperspectral remote sensing, data fusion, machine learning, and artificial intelligence.

Dr. Chanussot is a member of the Institut Universitaire de France from 2012 to 2017 and a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters). He is the founding President of the IEEE Geoscience and Remote Sensing French Chapter from 2007 to 2010, which received the 2010 IEEE GRS-S Chapter Excellence Award. He has received multiple outstanding paper awards. He was the Vice-President of the IEEE Geoscience and Remote Sensing Society, in charge of meetings and symposia from 2017 to 2019. He was the General Chair of the first IEEE GRSS Workshop on Hyperspectral Image and Signal Processing, Evolution in Remote sensing (WHISPERS). He was the Chair and the Co-Chair of the GRS Data Fusion Technical Committee from 2009 to 2011 and from 2005 to 2008, respectively. He was a member of the Machine Learning for Signal Processing Technical Committee of the IEEE Signal Processing Society from 2006 to 2008 and the Program Chair of the IEEE International Workshop on Machine Learning for Signal Processing in 2009. He is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON IMAGE PROCESSING, and the PROCEEDINGS OF THE IEEE. He was the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2011 to 2015. In 2014, he served as a Guest Editor for the *IEEE Signal Processing Magazine*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:53:14 UTC from IEEE Xplore. Restrictions apply.